

Benjamin Liu

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GitHub: <https://github.com/liuben10>

Skills

Web Frameworks: Spring MVC, Ruby on Rails, Express + Node.js, Django

Languages: Python, C++ (20+), Java and JVM languages (Kotlin, Scala), JavaScript with React and Flow

Cloud & Data: Kubernetes, Docker, Google Cloud (App Engine, Compute Engine, GCR, Functions, Datastore, Dataproc, Bigtable, Firebase), AWS (Lambda, EMR, EC2, S3, Athena, Kinesis), Cloudflare, Hadoop, Hive, MySQL, Spark, Kafka

Experience

August 2024 – May 2026

Coupage, Seattle – Staff Software Engineer, Price Determination (L6.1)

- Migrated 300+ asynchronous batch jobs from a team-specific cluster to a company-wide cloud platform, improving disaster recovery and adding standardized monitoring.
- Led remediation after a production security incident, designing and driving API security controls across 200+ Price Determination endpoints.[link](#)
- Modernized Price Determination infrastructure by upgrading platform services, libraries, and batch jobs from Spring 4 / Java 7 to Spring 6 / Java 17.
- Built controls to disable crawled-coupon price matching during competitor promotions, protecting an estimated \$1M+ in annual revenue.
- Owned platform-hardening and allowlisting tools for seller-management workflows, resolving critical issues surfaced during platform demos.
- Standardized time handling across Price Determination systems, establishing a consistent timezone for time-based data and reducing ambiguity across services.

December 2017 – April 2024

Facebook, Menlo Park / Seattle — Senior Software Engineer (IC5)

- Integrated Pacemaker, a C++ real-time spend estimator, with a new sharding API, strengthening disaster-recovery compliance, eliminating a hot-shard bottleneck, and improving scalability by 1.5×.
- Built a Python data-migration prototype for Billing and Revenue Infrastructure services using a shared backend, improving consistency and supporting Meta's digital sales tax obligations in Brazil.
- Led the migration of Advanced Analytics to CCPA- and GDPR-compliant datasets, reducing privacy and regulatory risk.
- Supported a multi-tenant clean-room environment hosting differentially private social datasets for Social Science One research on political advertising.
- Maintained a custom EMR clean-room environment for an Amazon-Facebook integration, enabling custom advertising lift experiments against Facebook data.

May 2016 – December 2017

Gear Launch, San Francisco — Full-Stack Software Engineer

- Took ownership of the print-on-demand marketing platform during a leadership transition, maintaining continuity of the core website.
- Launched support for mugs, creating an additional seasonal revenue stream.
- Built an Instant Shipping feature that bypassed batch-print processing, enabling an expedited shipping tier for advertisers.
- Integrated the product catalog with the Facebook Dynamic Ads API, enabling individual advertisers to use Facebook marketing tools.

June 2014 – May 2016

Xoom, a PayPal Service, San Francisco — Junior Software Engineer, Integrations

- Built a Spring Boot and Java REST API to automate event mapping, reducing manual work for production engineers.
- Built a transaction ETA prediction microservice using historical event-transition data.
- Developed and supported remittance integrations for 9 partners across 8 countries.

June 2013 – August 2013

Instinet — Software Engineer Intern

- Built an internal file-upload workflow for traders to submit documents.
- Maintained the internal website during the team lead's absence.

June 2012 – August 2012

Siemens Ultrasound — Software Engineer Intern

- Developed Python tests using waveform generators to validate ultrasound machines.

Education

2010–2014

UC Berkeley, Berkeley, CA — Electrical Engineering & Computer Science

B.S. in Electrical Engineering and Computer Science, emphasis in Computer Science; Minor in Mathematics